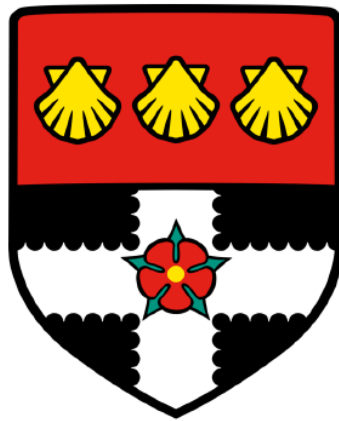


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DEPARTMENT OF METEOROLOGY



Improving Surface Gravity Wave Forecasts Using Data Assimilation

Owen Aljabar

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*A thesis submitted in partial fulfillment of the requirements for the degree of
Master of Science in Atmosphere, Ocean and Climate*

August 2024

Declaration of Authorship

I, Owen Aljabar, declare that this thesis titled, “Improving Surface Gravity Wave Forecasts Using Data Assimilation” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
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Abstract

Faculty Name
Department of Meteorology

Master of Science

Improving Surface Gravity Wave Forecasts Using Data Assimilation

by Owen Aljabar

The Thesis Abstract is written here (and usually kept to just this page). The page is kept centered vertically so can expand into the blank space above the title too...

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor...

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List of Symbols

a	distance	m
P	power	W (J s^{-1})
ω	angular frequency	rad

Chapter 1

Introduction

1.1 The need for Reliable Wave Forecasting

Offshore operations can only be carried out in sufficiently calm waters...

Chapter 2

Theory

2.1 Linear Wave Theory

Ocean waves are a distinctive visual feature of the ocean surface, and very important to offshore operations such as wind farms. They are an instance of surface gravity waves. This section will outline the linear wave theory relevant to this project, as derived in (Vallis, 2017), page 253.

Parameters describing surface gravity waves include the wavenumber k and direction θ , and the frequency ω . The first two describe the wave vector $\mathbf{k}$, which has magnitude k and direction θ . ω here denotes the absolute frequency measured in a fixed frame of reference. If the medium is subject to currents, this will differ from the relative frequency σ measured in a frame of reference moving with the mean current, due to the doppler effect. This difference is quantified by the equation

$$\omega = \sigma + \mathbf{k} \cdot \mathbf{U} \quad (2.1)$$

and σ and k are related by the dispersion relation

$$\sigma^2 = gk \tanh kD \quad (2.2)$$

The dispersion relation has two limiting forms:

- Deep water waves, where $kD \gg 1$, so $\tanh kD \approx 1$ and $\sigma^2 \approx gk$.
- Shallow water waves, where $kD \ll 1$, so $\tanh kD \approx kD$ and $\sigma^2 \approx gk^2D$

The phase speed is then given by

$$c = \frac{\sigma}{k} \approx \begin{cases} \sqrt{\frac{g}{k}} & \text{in deep water} \\ \sqrt{gD} & \text{in shallow water} \end{cases} \quad (2.3)$$

and the group speed may be expressed as

$$c_g = \frac{\partial \sigma}{\partial k} = \frac{1}{2}c \left(1 + \frac{2kD}{\sinh 2kD} \right) \approx \begin{cases} \frac{1}{2}c & \text{in deep water} \\ c & \text{in shallow water} \end{cases} \quad (2.4)$$

We can now use scale analysis to see that ω and σ generally have similar values. The ratio between $\mathbf{k} \cdot \mathbf{U}$ and σ in equation 2.1 is

$$\frac{\mathbf{k} \cdot \mathbf{U}}{\sigma} \sim \frac{kU}{\sigma} = \frac{U}{c}$$

Where $U = |\mathbf{U}|$ is the mean current speed. Since $c \gg U$ in general, this ratio is small and so $\omega \approx \sigma$.

Note that equations 2.1 - 2.4 are valid for linear waves of a single frequency and direction. Real ocean waves are more complicated than this, containing multiple frequencies propagating in multiple directions. The WAVEWATCH III model used in this project models ocean waves as a superposition of these simple linear waves.

Surface waves with multiple frequencies and directions are represented using a wave spectrum $F = F(\sigma, \theta)$, which is also a function of position and time. F represents the wave energy associated with a particular frequency σ and direction θ , so the total wave energy at any point in space and time is

$$E = \iint F d\sigma d\theta \quad (2.5)$$

Here, we also define the wave spectral moments which are derived from the unidirectional spectrum $S(\sigma) = \int F(\sigma, \theta) d\theta$:

$$m_k = \int S(\sigma) \sigma^k d\sigma \quad (2.6)$$

These moments can be used to derive other important wave variables, such as significant wave height and dominant wave period, from the spectrum (expand). Finally we define the wave action

$$N(\sigma, \theta) = \frac{F(\sigma, \theta)}{\sigma}$$

N is conserved in the absence of sources and sinks, making it the spectrum of choice for wave models.

2.2 Data Assimilation

As well as the model, we also have some direct observations of the waves which we would like to make use of to improve our predictions. We use techniques of data assimilation to achieve this.

We take a vector $\mathbf{x}$ to represent the state of the system, assuming it has a normal background distribution $\mathbf{x} \sim N(\mathbf{x}_b, \mathbf{B})$. We take a vector $\mathbf{y}$ to represent a set of observations, and assume $\mathbf{y}$ is related to $\mathbf{x}$ via an observation operator $\mathcal{H}$:

$$\mathbf{y} = \mathcal{H}(\mathbf{x}) + \varepsilon$$

where the error term ε is assumed to be normally distributed with mean zero and covariance $\mathbf{R}$: $\varepsilon \sim N(0, \mathbf{R})$. In our case $\mathcal{H}$ is linear i.e. there is a matrix $\mathbf{H}$ such that $\mathcal{H}(\mathbf{x}) = \mathbf{H}\mathbf{x}$, and so $\mathbb{P}(\mathbf{y}|\mathbf{x}) = N(\mathbf{H}\mathbf{x}, \mathbf{R})$. We can then express the distribution of $\mathbf{x}$ given some set of observations $\mathbf{y}$ using Bayes' theorem:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) = \mathbb{P}(\mathbf{y}|\mathbf{x}) \frac{\mathbb{P}(\mathbf{x})}{\mathbb{P}(\mathbf{y})} \quad (2.7)$$

This allows $\mathbb{P}(\mathbf{x}|\mathbf{y})$ to be determined since $\mathbb{P}(\mathbf{y}|\mathbf{x})$ and $\mathbb{P}(\mathbf{x})$ are known, and $\mathbb{P}(\mathbf{y})$ is independent of $\mathbf{x}$ so it acts as a normalisation constant. Since all the distributions involved are normal, $\mathbb{P}(\mathbf{x}|\mathbf{y})$ will be normally distributed also:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) \sim e^{-\frac{1}{2}(\|\mathbf{x}-\mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \|\mathbf{y}-\mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2)} \quad (2.8)$$

adopting the notation $\|\mathbf{a}\|_{\mathbf{M}}^2 = \mathbf{a}^\top \mathbf{M} \mathbf{a}$. Finding the most likely value of $\mathbf{x}$ given this distribution is equivalent to minimising a cost function

$$J(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2 \quad (2.9)$$

The state that minimises this cost function is called the analysis $\mathbf{x}_a$ and may be expressed as

$$\mathbf{x}_a = \mathbf{x}_b + \mathbf{K}\mathbf{d} \quad (2.10)$$

where $\mathbf{d} = \mathbf{y} - \mathbf{H}\mathbf{x}$ is the innovation and $\mathbf{K} = \mathbf{B}\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1}$ is the Kalman gain.

In our case, we have observations from multiple times $t = 0, 1, \dots, T$ spread out over some time window, and we expect the state $\mathbf{x}$ to evolve over this time window. This evolution may be predicted using the model as follows: let $\mathbf{x}_0$ represent the state of the system at $t = 0$, then we define the model operator $\mathcal{M}_t$ for each t such that $\mathbf{x}_{t+1} = \mathcal{M}_t(\mathbf{x}_t)$ as predicted by the model. The cost function then becomes

$$J(\mathbf{x}_0) = \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \mathbf{H}_t \mathbf{x}_t\|_{\mathbf{R}_t^{-1}}^2 \quad (2.11)$$

$$= \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \left\| \mathbf{y}_t - \hat{\mathbf{h}}_t \mathbf{x}_0 \right\|_{\mathbf{R}_t^{-1}}^2 \quad (2.12)$$

where the observations $\mathbf{y}_t$ and matrices $\mathbf{H}_t, \mathbf{R}_t$ are valid at time t , and $\hat{\mathbf{h}}_t = \mathbf{H}_t \mathcal{M}_{0 \rightarrow t}$ using the abbreviation $\mathcal{M}_{0 \rightarrow t} = \mathcal{M}_{t-1} \mathcal{M}_{t-2} \dots \mathcal{M}_0$.

We will use the 4DEnVar method, where a control variable transform is used to write $\mathbf{x}_0$ as a linear combination of the ensemble perturbations:

$$\begin{aligned} \bar{\mathbf{x}}_b &= \frac{1}{N} \sum_{n=1}^N \mathbf{x}_b^{(n)} \\ \mathbf{X}'_b &= \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_b^{(1)} - \bar{\mathbf{x}}_b, \mathbf{x}_b^{(2)} - \bar{\mathbf{x}}_b, \dots, \mathbf{x}_b^{(N)} - \bar{\mathbf{x}}_b \right) \\ \mathbf{x}_0 &= \bar{\mathbf{x}}_b + \mathbf{X}'_b \chi \end{aligned}$$

Here $\mathbf{x}_b^{(n)}$ represents the background state from ensemble member n , and N is the ensemble size. The cost function can then be rewritten in terms of χ and simplified by linearising $\hat{\mathbf{h}}_t$ around $\bar{\mathbf{x}}_b$:

$$J(\chi) = \frac{1}{2} \|\chi\|^2 + \frac{1}{2} \sum_{t=0}^T \left\| \mathbf{y}_t - \hat{\mathbf{h}}_t \bar{\mathbf{x}}_b - \hat{\mathbf{H}}_t \mathbf{X}'_b \chi \right\|_{\mathbf{R}_t^{-1}}^2 \quad (2.13)$$

where $\hat{\mathbf{H}}_t = \mathbf{H}_t \mathbf{M}_{0 \rightarrow t}$ is the linearisation of $\hat{\mathbf{h}}_t$. This cost function is simplified by taking a background error covariance matrix of the form $\mathbf{B} = \mathbf{X}'_b \mathbf{X}'_b{}^\top$, which reduces to the identity under the control variable transform. This cost function has gradient

$$\nabla J(\chi) = \chi + \sum_{t=0}^T \left(\hat{\mathbf{H}}_t \mathbf{X}'_b \right)^\top \mathbf{R}_t^{-1} \left(\hat{\mathbf{h}}_t \mathbf{X}'_b \chi + \hat{\mathbf{H}}_t \bar{\mathbf{x}}_b - \mathbf{y}_t \right)$$

A major advantage of this formulation is that we can make the approximation

$$\mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right) \approx \mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} \right) - \overline{\mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b \right)}$$

Since $\mathbf{H}_t \mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right)$ is the n^{th} column of $\hat{\mathbf{H}}_t \mathbf{X}'_b$, this approximation allows us to compute the cost function and its gradient without a tangent linear or adjoint model, which is a computationally expensive aspect of most variational data assimilation schemes (Bannister, 2017).

Chapter 3

Experimental Setup

3.1 Model Description and Domain

Waves are modelled by Fugro using WAVEWATCH III, which uses as its base spectrum a wavenumber-direction spectrum $F(k, \theta)$, rather than the frequency-direction spectrum described above. The spectrum is discretised using 29 sampled wavenumbers k and 24 angles θ . The wavenumber-direction spectrum is related to the frequency-direction spectrum by the Jacobian transform

$$F(\sigma, \theta) = \frac{\partial k}{\partial \sigma} F(k, \theta) = \frac{1}{c_g} F(k, \theta)$$

Wave propagation is then described by the equation

$$\frac{D}{Dt} N(k, \theta) = \frac{S}{\sigma} \quad (3.1)$$

Where $N(k, \theta) = F(k, \theta)/\sigma$ as before and S describes the net effect of sources and sinks for the spectrum $F(k, \theta)$.

In deep water, S consists of three main parts: an atmospheric interaction term S_{in} , a nonlinear wave-wave interaction term S_{nl} and a dissipation term S_{ds} .

S_{in} is dominated by an exponential wind-wave growth term, so S_{in} is equal to some coefficient multiplied by $\sigma N(k, \theta)$. This leads to a feedback which is usually positive but can be negative in the case of swell, and represents the main source of wave energy.

S_{ds} is generally dominated by wave breaking, and it represents a sink of wave energy. Similar to S_{in} , it is equal to a negative coefficient multiplied by $\sigma N(k, \theta)$. The balance between S_{in} and S_{ds} determines the overall growth or decay of wave energy.

S_{nl} represents interactions between waves due to nonlinear processes, which results in energy transfer between waves of different frequencies. (maybe expand - Janssen, 2004)

In shallow water, processes such as wave-bottom interactions, wave breaking and reflection off shorelines become important, so additional source terms are included to describe this.

From the forecasted spectra, other variables such as wave height, period and mean direction are derived. The three variables we will be focusing on in this study are significant wave height H_s , and two measures of the dominant wave period T_{m01} and T_{m02} . These are related to the spectrum via the spectral moments defined in 2.6 as follows:

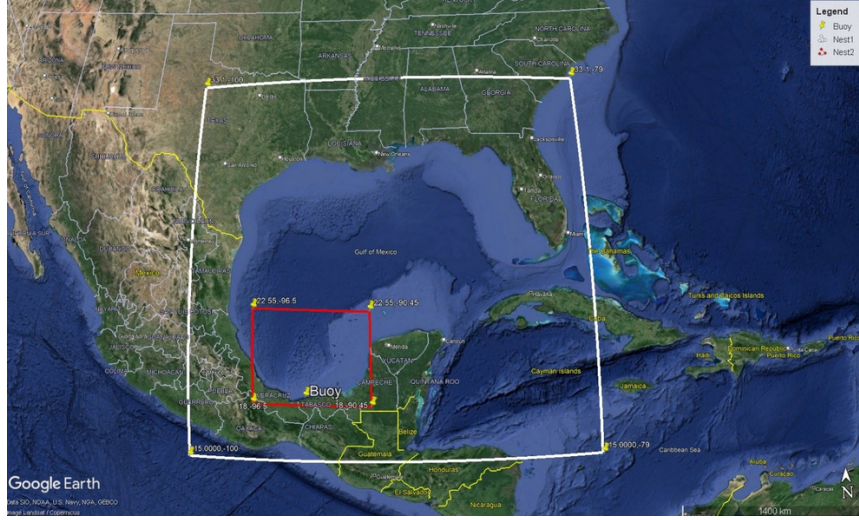


FIGURE 3.1: Model domains shown on map.

$$H_s = 4\sqrt{m_0} \quad (3.2)$$

$$T_{m01} = \frac{m_0}{m_1} \quad (3.3)$$

$$T_{m02} = \sqrt{\frac{m_0}{m_2}} \quad (3.4)$$

As forcing, WAVEWATCH III uses just the 10 metre wind, and this comes from the GEFS ensemble. The GEFS forecast contains one control run plus 20 perturbed ensemble members. Each GEFS run acts as forcing for one member of the wave ensemble. (Do they?)

The domain in which forecasts are produced lies between $15.0 - 33.0^\circ\text{N}$ and $79.1 - 100^\circ\text{W}$ in the gulf of Mexico. Fugro use a nested domain to produce the forecasts, running WAVEWATCH III on a 0.5° resolution globally rectangular grid, and two higher resolution nested grids covering the region of interest. There is a 0.1° resolution grid spanning the region described above, and a 0.05° grid spanning from $18 - 22.5^\circ\text{N}$ and $90.5 - 96.5^\circ\text{W}$. These domains are shown on a map in figure 3.1.

Forecasts are produced at 12Z each day, and run up to seven days ahead. We have seven of these daily runs available, with start dates spanning from 06/03/2022 to 12/03/2022. (include schematic) As we do not have access to the model code in this project, we will treat each of these runs as one 'timestep' of the DA system. Further description of how this works can be found in section 3.4.

The available model runs give forecast data for a period spanning from 12Z on 06/03/2022 to 19/03/2022. The main feature of this period is a large swell event with north-northwesterly waves, associated with a cold front passing through the domain. This event is illustrated in figure 3.2.

3.2 ADCP Observations

Our observations come from an acoustic doppler current profiler (ACDP) buoy, which estimates the spectrum by measuring the wave induced velocity and surface height

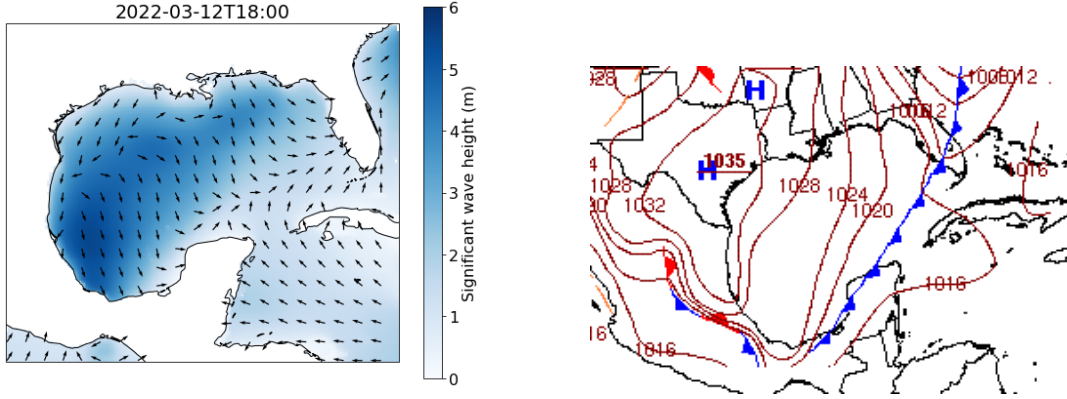


FIGURE 3.2: Left: Significant wave height (colours) and direction (arrows) in the domain at 18Z on 12/03/2022. Right: Surface analysis chart valid for 18Z on 12/03/2022, taken from NOAA WPC surface analysis archive

variations. According to Vallis, 2017, the velocity and surface height anomaly induced by a simple wave with wave vector $\mathbf{k} = (k_x, k_y)$, frequency σ and amplitude a is

$$\begin{aligned} u &= a \frac{k_x}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ v &= a \frac{k_y}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ w &= -ia \frac{k}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \sinh k(z + D) \\ \eta &= a \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \end{aligned}$$

where only the real part of each expression should be taken. Note that the horizontal velocity (u, v) is always in the same direction as the wave vector $\mathbf{k}$. From the ADCP measurements, a timeseries of wave induced energy per unit area $E = \int_{-D}^{\eta} \frac{1}{2} \rho_0 \|\mathbf{u}\|^2 dz + \frac{1}{2} \rho_0 \eta^2$ can be calculated. This is equivalent/a good approximation to the wave energy in equation 2.5 (source) so a Fourier transform of E gives the wave spectrum integrated across all directions $E(\sigma) = \int F(\sigma, \theta) d\theta$ (how is directional distribution estimated?)

Other variables are then deduced from this spectrum, including H_s , $Tm01$ and $Tm02$ as described above.

We have observations available over a roughly 6 month period spanning from October 2021 to April 2022, which includes the period for which forecasts are available. A timeseries of the observed significant wave height from the forecast period is included in figure 3.3. The peak in wave height late on the 12th of March is clearly visible on the plot.

To determine the magnitude of the observation errors, several methods are available. A simple one is to plot a histogram of observation - ensemble mean, called an $o - b$ plot. The spread in $o - b$ comes from a combination of observationa and background error, so a simple way of estimating the observation error is to assume that observation error accounts for a constant proportion of this error, say 50%. An $o - b$ histogram for significant wave height is shown in figure 3.4. From the plot we see that the spread on the histogram is approximately 1m, so the standard of observational errors in significant wave height may be estimated to be approximately 50cm.

This approach may be quantified more precisely following Desroziers et al., 2006...

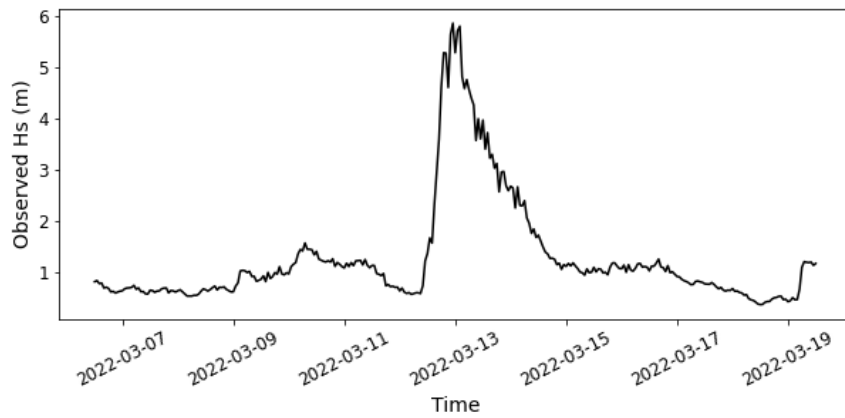


FIGURE 3.3: Observed timeseries of significant wave height from ADCP buoy

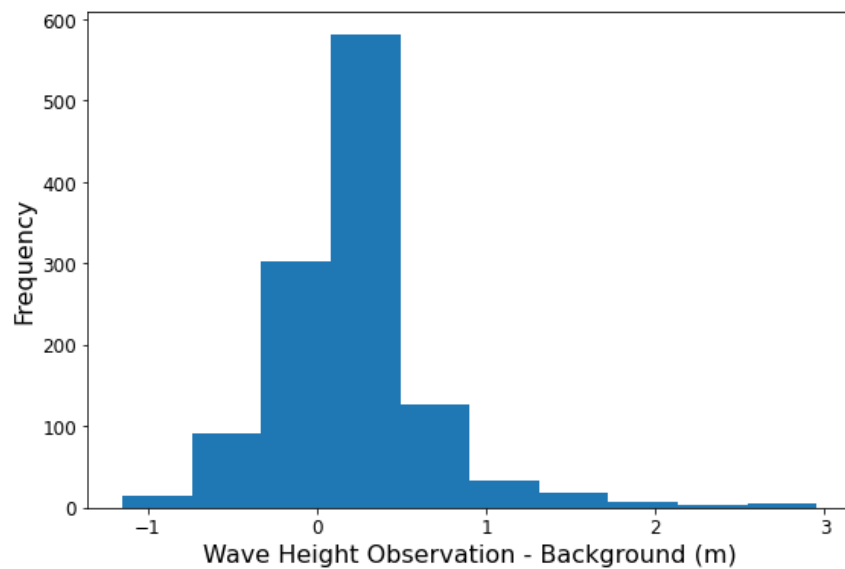


FIGURE 3.4: Observation - background histogram for significant wave height.

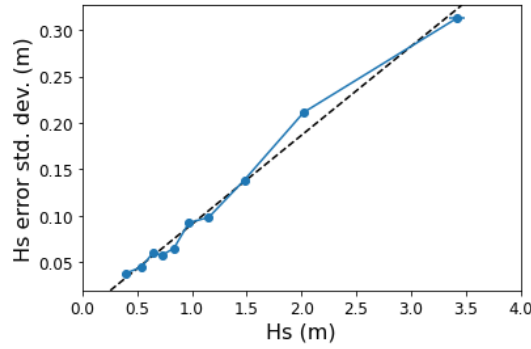


FIGURE 3.5: A plot of Hs error standard deviation against Hs.

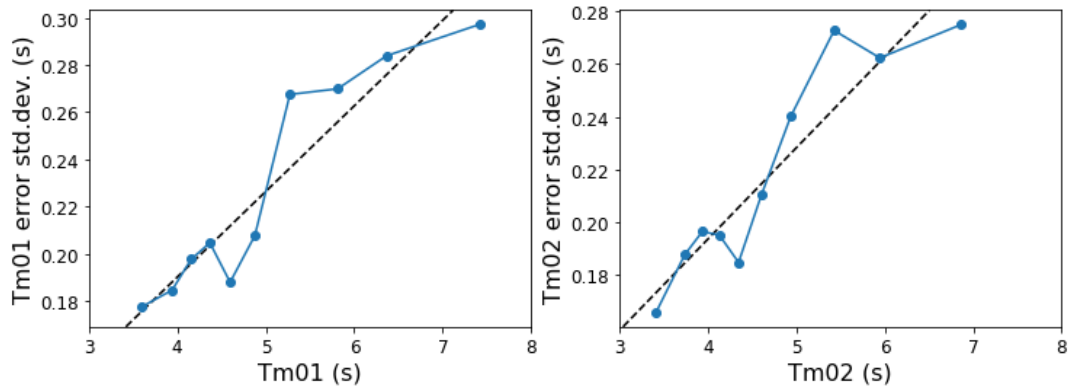


FIGURE 3.6: Plots of Tm01 and Tm02 error standard deviation against Tm01 and Tm02 respectively.

Another available method is that used in Houghton et al., 2023. They used a colocation study to estimate the error in wave height observations from a distributed network of drifting sofar spotter buoys. Pairs of simultaneous observations from locations within 50km of each other were collected and binned according to the average wave height of each pair. The difference of each pair was taken, and the standard deviation of of differences from each bin was computed. They found a strong dependence of error on wave height.

In our case, we only have observations from one buoy and they are available hourly, so we take the pairs to be observations one hour apart. We then calculate the standard deviation using the method described above to obtain the plot in figure 3.5. As in Houghton et al., 2023, we find a strong linear relationship between errors and wave height, with the error standard deviation being approximately 9.6% of the wave height. Repeating this process with the periods Tm01 and Tm02 yields the plots shown in figure 3.6.

The error distributions for Tm01 and Tm02 also fall approximately on straight lines, though the fit is not as good as for Hs. The gradients are also shallower, with the error standard deviations being approximately 3.5% of the observed variable for both. So we will model the error standard deviation in Hs as being equal to 10% of Hs, and the error standard deviation in Tm01 and Tm02 as 3.5% of Tm01 and Tm02 respectively. (maybe bump up - why?)

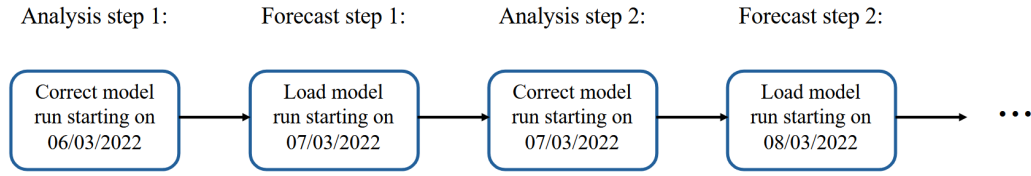


FIGURE 3.7: Schematic of the forecast and analysis steps from the ETKF scheme employed in this project.

3.3 Twin Experiment

In this study, we will not be using real observations in our assimilation but setting aside one ensemble member to represent the 'truth' and generating fictional observations based on this forecast data. This is known as a twin experiment, and has several advantages over assimilating real observations:

- The 'truth' is available away from the location of observations, so forecast error can be calculated over the whole model domain. This allows the effectiveness of DA to improve forecasts away from observation locations to be evaluated

The observations are generated by sampling values from the truth at predefined observation locations and times. Here we sample hourly values from the nearest gridpoint to the buoy to match the real observations that are available. To represent observation error, some random noise is added to the sampled values. This noise is drawn from a normal distribution with standard deviation equal to the error standard deviation derived in section 3.2

3.4 Data Assimilation Setup

To perform the data assimilation we use pyPDAF, the python interface for the fortran-based Parallel Data Assimilation Framework (PDAF). (cite Yumeng?) We will use the ETKF method which comes builtin with PDAF. As we are not able to run the WAVEWATCH III model ourselves, at each analysis step we will correct one 7-day forecast window all at once, and then the forecast step will consist of selecting data from the next 7-day window. In this way the time within the 7-day window is treated like another spatial coordinate by the DA system. A schematic of this process is included in figure 3.7

At each analysis step, we will assimilate observations from the first three days of the forecast window. This allows us to investigate errors at times after the assimilation window and assess the capability of DA to improve the forecast at future time when observations are unavailable.

Appendix A

Frequently Asked Questions

A.1 How do I change the colors of links?

The color of links can be changed to your liking using:

```
\hypersetup{urlcolor=red}, or  
\hypersetup{citecolor=green}, or  
\hypersetup{allcolor=blue}.
```

If you want to completely hide the links, you can use:

```
\hypersetup{allcolors=.}, or even better:  
\hypersetup{hidelinks}.
```

If you want to have obvious links in the PDF but not the printed text, use:

```
\hypersetup{colorlinks=false}.
```


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